

Mixed Mechanics

Selecting the right model

YOUR AIM TODAY

Choose between SUVAT, forces, energy, power and momentum.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

The ideas you need

The question will not name the method. Identify what is conserved or what stays constant.

- Constant acceleration plus u, v, a, s, t suggests SUVAT.
- For forces and acceleration, draw a diagram and use $F=ma$.
- Heights and speeds often suggest energy.
- Short impacts suggest impulse or momentum.
- Rates of work suggest power.

MEMORY RULE

List known quantities and the requested quantity before selecting an equation.

Selection

WORKED STEPS

Speed after falling through a height: energy is shortest. Speed after a stated time at constant acceleration: SUVAT.

Verification

WORKED STEPS

After solving, check units, sign and physical size.

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. From rest, $a=3 \text{ m/s}^2$ for 5 s. Find distance.	
2. A 4 kg block has 18 N forward and 6 N resistance. Find acceleration.	
3. Find speed after falling 5 m from rest.	
4. A 2 kW engine works for 30 s. Find work.	
5. A 3 kg particle at 4 m/s receives impulse -18 Ns. Find final velocity.	
6. A 5 kg block on a table connects to 5 kg hanging, smooth. Find acceleration.	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	$s=0.5at^2=37.5 \text{ m}$.
2	Resultant 12 N; $a=12/4=3 \text{ m/s}^2$.
3	$v=\sqrt{2gh}=9.90 \text{ m/s}$.
4	$W=Pt=2000(30)=60000 \text{ J}$.
5	$3(v-4)=-18$, so $v=-2 \text{ m/s}$.
6	$5g=10a$, so $a=g/2=4.9 \text{ m/s}^2$.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.